

Validation Harness for Synthetic Benchmarks

Project ARIA Test Harness

TABLE I
MANIPULATION BENCHMARK ACROSS 10 STANDARDIZED TASKS

Model / Architecture	Params	Eval. Setting	Overall Success (%), Wilson 95% CI	Mean Execution (s)	VRAM (GB)	Inference (ms)
OpenVLA	7B	Physical (matched)	68.0% [58.3, 76.4]	5.1 ± 1.2	16.5	140
LeRobot ACT	80M	Physical (matched)	72.0% [62.5, 79.9]	3.8 ± 0.6	6.0	35
Project ARIA (Ours)	–	Physical ($N = 20$)	85.0% [64.0, 94.8]	3.3 ± 0.5	7.2	18

TABLE II
MULTI-SPEED CONVEYOR SORTING SWEEP ACROSS SIX BELT
VELOCITIES, WITH WILSON 95% CIs

Belt Speed (v_{belt})	Cycles	Grasp Succ.	Sort Succ.	Overall Rate (%)	Wilson 95% CI
0.02 m/s	4	4	4	100.0%	[51.0, 100.0]
0.04 m/s	4	4	4	100.0%	[51.0, 100.0]
0.06 m/s	4	4	4	100.0%	[51.0, 100.0]
0.08 m/s	4	3	3	75.0%	[30.1, 95.4]
0.10 m/s	4	2	2	50.0%	[15.0, 85.0]
0.12 m/s	4	1	1	25.0%	[4.6, 69.9]

I. INTRODUCTION

Testing synthetic table code generation for the Project ARIA validation campaign.

The quantitative benchmarks are compiled in Table I, Table II, and Table III.

TABLE III
ARCHITECTURE-LEVEL ABLATION STUDY ACROSS SIX MODULAR AND SAFETY CONFIGURATIONS ($N = 60$ TRIALS PER CONFIGURATION)

Configuration / Pipeline Variant	Trials	Success Rate (%), Wilson 95% CI	Mean Latency (ms)	Tail Latency p_{95} (ms)	Recovery Time (ms)	Unsafe Plans Intercepted
Monolithic Sequential	4	50.0% [15.0, 85.0]	461.2	510.0	–	–
Modular Synchronous	4	75.0% [30.1, 95.4]	307.5	340.0	–	–
Modular Asynchronous (Full ARIA)	4	100.0% [51.0, 100.0]	182.5	190.0	–	4
ARIA w/o Lifecycle Recovery	4	75.0% [30.1, 95.4]	199.2	245.0	–	3
ARIA w/o ReachabilityAgent	4	50.0% [15.0, 85.0]	195.8	225.0	–	–
ARIA w/o SafetyAgent / HITL	4	50.0% [15.0, 85.0]	187.0	215.0	–	–